

Internet Relay Chat Protocol

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Abstract

This memo describes the communication protocol for an IRC-style client/server system for the Internetworking Protocols class at Portland State University.

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Introduction

This specification describes a simple Internet Relay Chat (IRC) protocol by which clients can communicate with each other. This system employs a central server which "relays" messages that are sent to it to other connected users. Users can join rooms, which are groups of users that are subscribed to the same message stream. Any message sent to that room is forwarded to all users currently joined to that room. Users can also send private messages directly to other users.

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Conventions used in this document

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119 [RFC2119].

In this document, these words will appear with that interpretation only when in ALL CAPS. Lower case uses of these words are not to be interpreted as carrying significance described in RFC 2119.

In this document, the characters ">>" preceding an indented line(s) indicates a statement using the key words listed above. This convention aids reviewers in quickly identifying or finding the portions of this RFC covered by these keywords.

Basic Information

All communication described in this protocol takes place over TCP/IP, with the server listening for connections on port 7734. Clients connect to this port and maintain this persistent connection to the server. The client can send messages and requests to the server over this open channel, and the server can reply via the same. This messaging protocol is inherently asynchronous - the client is free to Marohn Expires June 1, 2016 [Page 3] Internet-Draft Internet Relay Chat Class Project December 2015 send messages to the server at any time, and the server may asynchronously send messages back to the client.

As is described in [4.2], both the server and client may terminate the connection at any time for any reason. They MAY choose to send an error message to the other party informing them of the reason for connection termination.

The server MAY choose to allow only a finite number of users and rooms, depending on the implementation and resources of the host system. Error codes are available to notify connecting clients that there is currently a high volume of users or groups accessing the server.

Message Infrastructure

Generic Message Format